Entangled Itineraries

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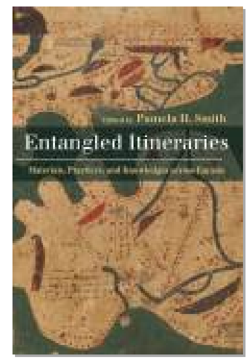

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Chapter 2

Trans-Eurasian Routes of Exchange

A Brief Historical Overview

Tansen Sen and Pamela H. Smith

Long-term and long-distance flows of goods, knowledge, texts, practices, and peoples (material complexes) moved back and forth across trans-Eurasian paths of exchange—the "Afro-Eurasian ecumene," as Ptolemy conceived of this expanse, a conception followed in the visualizations of Eurasian mapmakers such as Muhammed al-Idrisi (c. 1100–1165), in his map made in service of Roger II of Sicily in 1154 (figure 2.1), and fifteenth-century German cartographers (figure 2.2).[1] From at least the second millennium BCE, the Eurasian world was integrated through the movement of peoples and the diffusion of ideas and goods. This integration took place through the routes taken by materials, practices, and knowledge, and such routes can be more important than their roots. In the same vein, Arjun Appadurai has argued that "bumps and blocks, disjunctures and differences are produced by the variety of circuits, scales and speeds which characterize the circulation of cultural elements."[2]

Appadurai explains this phenomenon through the frameworks of "the circulation of forms and the forms of circulation." By "forms," Appadurai indicates "a family of phenomena, including styles, techniques, or genres, which can be inhabited by specific voices, contents, messages, and materials." This "circulation of forms," he contends, "produces new and distinct genre experiments, many of which are forced to coexist in uneven and uneasy combinations." Associated with this circulation of forms is what Appadurai calls the "work of the imagination," what "humans do as they strive to extend their chances of survival, improve their horizons of possibility, and increase their wealth and security." Moreover, these activities also produce "locality." The local "was not just an inverted canvas on which the global was written, but . . .

Figure 2.1. Al-Idrîsî, Nuzhat al-mushtaq fî ikhtirâq al-âfâq, map of the world, from a copy made in 1456 by 'Alî ibn Hasan al-Hûfî al-Qâsimî in Cairo and now preserved at the Bodleian Library, Oxford, as MS. Pococke 375, fols. 3v–4. Also called the Kitab Rujar (Book of Roger of Sicily; the Tabula Rogeriana; 1154). Public domain.

itself was a product of incessant effort."[3] With regard to the relationship between the global and the local, Appadurai suggests we "accept that the global is not merely the accidental site of the fusion or confusion of circulating global elements. It is the site of the mutual transformation of circulating forms." These transformations, he explains, take place through the "work of imagination," which in turn produces locality.[4] Although Appadurai's study relates to issues of contemporary globalization and the resulting connections and circulations, the frameworks of the "circulation of forms" and the "forms

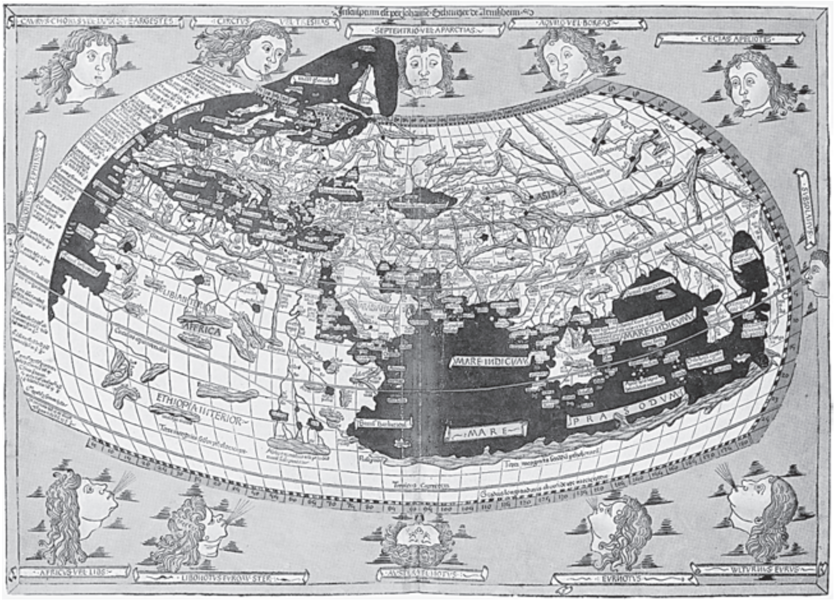

Figure 2.2. Claudius Ptolemy, *Oecumene,* Johannes Schnitzer, engraver (1482). Public domain.

of circulation" could, as he acknowledges, be applied to the earlier phases of cross-cultural connections.

Two other conceptual frameworks are useful in understanding the movement of materials across Eurasia and the integration this achieved. Like Appadurai, Homi Bhaba underscores the significance of disjunctures and differences during circulations, but he also stresses the importance of convergence: "Circulation takes a measure of mobility—the movement of languages, ideas, meanings, cultural forms, social systems—as it converges in specific and singular spaces of representation negotiated through a dialogue of difference. Incommensurable customs, disjunctive symbolic structures, itineraries that are diverse and yet proximate, continuities that become contingent over time—these dis-proportionate convergences generate an energy of interdisciplinary circulation." Convergence, for Bhaba, "is not about a practice or project as an end in itself, even if that end is an entangled encounter of diverse thoughtways and institutional intersections. The aim of convergence as critique is to track the spatial and temporal territories that open up within, and through, the act of circulation. The iterative dynamics of circulation and convergence reveal lateral meanings and interstitial spaces produced *in transit.*"[5] Bhaba's emphasis on the "act of circulation" emphasizes again the importance of following itineraries. In this he agrees with the notion of circulatory history put forth by Prasenjit Duara. Circulatory history, according to Duara, is a mode "in which ideas, practices and texts enter society or locale as one kind of thing and emerge from it considerably transformed to travel elsewhere even as it refers back, often narratively, to the initiating moment."[6]

As these scholars make clear, itineraries should be conceptualized as neither random nor unidirectional. Rather, already at an early date, several regions of the Eurasian world were connected through the networks of exchange that entailed deliberate, repeated, and multidirectional movements. Below are some examples of people, objects, and ideas that traveled across Eurasia and contributed to the integration of the geographical spaces and terrains, triggered various forms of circulations, and led to instances of transformation and convergence.

Early Eurasian Connections

Several millennia before the Common Era, the seafaring community of Austronesians and the nomadic tribes who inhabited the Eurasian steppe zone connected far-flung areas (including parts of Africa) through their networks of commodity exchange and seasonal migrations.[7] While the former group, originating from the South China Sea region, used their canoes to travel across the Indian Ocean and introduced new styles of pottery-making into places such as southern India, the Indo-European nomads are credited with the circulation of expertise associated with bronze-making, the use of composite bows, horse riding, and iron stirrups.[8] The movements and contributions of Austronesians and Indo-Europeans clearly indicate the shortcomings of conceptualizing or tracking the itineraries of peoples, goods, and ideas based on either continental designations or political boundaries. For example, the itinerary of Maldivian cowries demonstrates the lack of separation between Europe and Asia: from the second millennium BCE through the second millennium CE, a considerable volume of cowries from the Maldives Islands in the Indian Ocean traveled in several directions, to the west into the African mainland and Europe and to the east across the Bay of Bengal into eastern India, Myanmar, and all the way to the Yellow River valley of China as early as the second millennium BCE.[9]

Although there are no textual records of the early trade, it is apparent from tracing the itineraries of these objects that the relaying of the commodity involved diverse modes of transportation over a varied terrain—mountainous tracks, river crossings, and overland pathways—and several different groups of traders. The itinerary of cowries from the Maldives Islands to the central plains of China may have been relatively fixed, as archaeological findings in Assam, Burma, and Yunnan help identify some of the relay sites that formed part of the networks of cowrie trade.[10] From the Maldives Islands, the cowries reached ports in present-day Bengal and then were transported by overland routes through Assam and Burma into the Yunnan region where they were used as monetary units. Some may have also reached the Burmese ports first and then moved along land routes into Yunnan. From Yunnan, cowries were sent on to the Yellow River valley and to the Shang capital, Anyang.

These earlier networks of exchange served as the foundations for a deeper integration of the Eurasian world during the last five centuries of the first millennium BCE. The formation of empires, the emergence of urban centers, and a better understanding of monsoon winds in the Indian Ocean were some of the other key factors stimulating intra-Eurasian interactions during this period. The Achaemenid Empire (550–330 BCE) and the Hellenistic polities that spanned from the Indus to the Aegean Sea, the Mauryan Empire (322–185 BCE) in South Asia, and the Han Empire (206 BCE–220 CE) in China, all contributed to the building of roads and bridges that facilitated the movement of peoples and goods. The growth of local economies and the increase in foreign trade led these empires to experiment with standardized monetary and weighing systems, which encouraged further expansion of trade. The increase in population and the appearance of cities created demand for local and foreign staple goods and rich luxuries. It was during this period that the corpora of knowledge were codified, first orally and subsequently in texts, in transregional languages—Greek, Chinese, Sanskrit, and early Persian—that came to underpin the construction of remarkably durable fictive identities, ostensibly knitting together diverse peoples into cross-regional groupings, imagined at various times and places as homogenous; unified by politics, bureaucracy, culture, race, or ethnicity.

Parthian traders, Greek and Roman sailors, and South and Southeast Asian merchants connected the ports and markets of the Eurasian world. Roman glass, Chinese silk, precious and semiprecious stones and beads from Central and South Asia, and Southeast Asian gold and tin became some of the key commodities in the increasingly lucrative cross-regional commerce. Western Han officials based in the Hepu region (in present-day Guangxi province of China) owned and were buried with their Roman glassware, jewelry made of imported beads and precious and semiprecious stones (such as carnelian, agate, and lapis lazuli), and pearls originating from various regions of South Asia.[11] Glass objects from the Roman Empire moved through ports in Southeast and South Asia, entering South Asia through the commercial networks that connected the Coromandel coast in southwest India to the ports in the Red Sea. From the Coromandel coast, the imported glassware passed through Southeast Asia and reached Jiaozhi, Hepu, and Guangzhou on the southern frontiers of the Qin and Han dynasties of China.[12]

The main consumers of these foreign luxuries in China were members of the elite who had settled in the frontier regions after the southward expansions by the two empires. The beads and semiprecious stones excavated from the Hepu tombs mostly originated in South Asia. However, as some scholars have suggested, it is possible that foreign artisans based in Southeast Asia were responsible for reprocessing and even manufacturing these items according to South Asian style for local consumption and for export to China.[13] This practice of customizing commodities for foreign and local consumption was also common in long-distance commercial interactions. The oasis regions of Central Asia are known to have tailored and redesigned silk fabrics originating in China according to the requirements of markets elsewhere. Such customized Chinese silk products were transported to foreign ports and markets through South Asia. The first-century Greco-Roman work that describes navigation and trade around the Roman world, entitled *The Periplus of the Erythraean Sea*, mentions Chinese silk yarns and silk fabrics as some of the commodities exported from ports known as Barbarikon (present-day Karachi, Pakistan) and Barygaza (present-day Bharuch in Gujarat, India).[14]

The procurement of foreign goods for domestic consumption and transit trade gradually increased over time and apparently dominated long-distance commerce after the tenth century. Horses, porcelain, and silver were some of the commodities that circulated widely within the Afro-Eurasian world and contributed significantly to the integration of far-flung ports and towns through networks of commercial exchanges. Horses suitable for warfare were in high demand by Chinese courts from the Han dynasty. Central Asian and Persian horses, and later Tibetan ponies and mounts, were imported in large numbers throughout most of China's imperial history. South Asia was an important part of this supply chain. Arabian horses were first exported to the Malabar coast, and from there they were transshipped to China. During the twelfth and thirteenth centuries merchants from Kish in the Persian Gulf played an important role in supplying these Arabian horses to South Asia.[15] The lucrative nature of the trade in horses is underscored in the works of the Persian historian Waṣṣāf and the Italian traveler Marco Polo, who mentions that "some, indeed most of them, fetch fully two hundred pounds of Touraine apiece." Waṣṣāf reports that fourteen hundred horses were sold to the Pandyan ruler named Sundara on the Malabar coast by Arab traders at the price of 220 dinars of red gold each, including for those that might be lost at sea. Waṣṣāf also notes that ten thousand horses were exported annually from the Persian Gulf during the reign of Atabeg Abu Bakr (r. 1226–1260) to ports in South Asia. The revenue from such exports, according to him, amounted to 2,200,000 dinars.[16] The continuation of this transit route two centuries later is confirmed by Fei Xin, who traveled to the Malabar coast (and beyond) with the Ming admiral Zheng He (1371–1433) in the early fifteenth century. According to Fei Xin, who accompanied Zheng He on some of the voyages, horses were brought to the Malabar coast from West Asia, bred locally, and "transferred" for hundreds or thousands of coins for onward sale to Ming China.[17]

Another trading circuit through which horses were supplied to markets in China is reported by the thirteenth-century Persian-language record from South Asia known as *Tabaqāt-I Nāsirī*. According to the work, the trade in horses was carried out by the Tibetans in Bengal and Kamarupa (present-day Assam state of India). In the market town of Lankhnauti in northern Bengal, according to this work, about fifteen hundred horses were sold every morning.[18] Some of these horses, it seems, were exported to China through the maritime routes that connected the Bay of Bengal ports to the coastal regions of China. Ming sources mention that representatives from Bengal often presented horses as tribute to the court. These representatives, in addition to presenting tribute, must have also sold some of the horses in local markets or at least informed the Ming court about their availability in Bengal. Since Bengal did not produce horses, the gifts brought to Ming China were most likely procured in the markets of Lankhnauti. This circuit also featured the export of silver from the mines in China to Bengal during the precolonial period. Then, with Spanish colonization and the discovery of silver mines in the Americas, the webs of commerce expanded significantly, connecting regions such as Yunnan, Sichuan, Tibet, Bengal, Assam, and Myanmar to the Indian Ocean on one end and to the Pacific on the other.

The trade in this circuit continued into the twentieth century, with significant ramifications: since at least the Song period, the Tibetans had procured Chinese tea in exchange for horses sold on the borderlands of Sichuan and Kham. Because of political instability in the Sichuan-Kham region in the early twentieth century, the trade in tea expanded geographically to include Myanmar and British India. Tea from the Yunnan region reached Lhasa through Myanmar and north Bengal and Sikkim. The merchants engaged in this trade often went to Calcutta to remit their earnings to China, using the remittance services of the overseas Chinese and governmental financial institutions, such as the Indian branches of the Bank of China. As a result, urban centers such as Calcutta and Kunming became important parts of the web of commerce, as providers of financial services rather than simply as markets for commodity exchange.[19]

During the Han period, the envoy Zhang Qian (d. 133 BCE) gave the first written account of the trade through this Yunnan-Bengal region. "When I was in Daxia [Bactria]," he writes, "I saw bamboo canes from Qiong and cloth made in the province of Shu [Sichuan]. When I asked the people how they had gotten such articles, they replied, 'Our merchants go to buy them in the markets of Shendu [India].'"[20] This is just one of numerous records of political and commercial contact between China and western Asian lands to be found in the *Hou Han Shu*, the official history of the Chinese Han dynasty.[21] Indeed, as Andre Gunder Frank notes, "when we lack such written official testimony by contemporaries or archeological evidence, we may suppose that trade was still older and more widespread than recorded. . . . That is, trade relations preceded and exceeded diplomatic ones."[22] Key nodes for such trading activity, including Babylonia in Persia and Arikamedu in southern India, formed throughout the Eurasian world. The commercial linkages also triggered the movement of artisans and the diffusion of technical and geographical knowledge.[23] Itinerant traders also helped spread of religious ideas, images, and artifacts, including those associated with Buddhism (see Sen, this volume).

In the first century CE the founding of the Kuṣāṇa Empire, which extended from Central Asia to the Gangetic plains of India, facilitated the interweaving of commercial activity and religious transmissions across the overland and maritime routes. The Sogdian traders used the Kuṣāṇa territories to establish a vast network that, for example, supplied horses to the Chinese markets and procured silk in return. The Sogdians were also some of the earliest transmitters and translators of Buddhist doctrines entering

China.[24] Persian, Sri Lankan, and Southeast Asian sailors also contributed to the spread of Buddhism by ferrying missionaries and pilgrims around within an emerging Buddhist cosmopolis. The spread of Buddhism to foreign lands resulted in the demand for related ritual objects and other religious paraphernalia that also benefited merchant communities. This intimate association between merchant communities and the spread of Buddhist doctrines persisted throughout most of the first millennium CE.[25]

Like Buddhism, the spread of Islam from the seventh century CE triggered the circulation of religious paraphernalia, knowledge and information, art and architecture. Also comparable to Buddhism, but with more intensity, Islam prompted the movement of missionaries and pilgrims. The emphasis on Hajj as one of the pillars of the faith resulted not only in frequent travels of people from different parts of Eurasia to Mecca but also in the development of transportation facilities and an intensification of economic transactions associated with pilgrimage activities.[26] The post-seventh-century period also witnessed an unprecedented growth in maritime commerce. The networks of Muslim traders belonging to several different ethnic groups and transformations in navigational and shipbuilding technologies were some of the contributing factors in the upsurge of trade across the Indian Ocean. The ninth-century Belitung shipwreck is a prime example of this growth in maritime connections. This Arab dhow, which sailed from the Chinese coast and sank in the Indonesian waters near the island of Belitung, carried over sixty-four thousand pieces of Chinese porcelain ware, gold and silver objects, and many other objects. These items seem to have been commercial goods as well as gifts for foreign rulers.[27] Archaeological finds have demonstrated that Chinese ceramics traveled as far as East Africa during this period.[28] In fact, it was during almost the same period that a Chinese envoy from the Tang court is reported to have traveled to the Abbasid Empire.[29] It is evident that maritime routes had developed into important conduits for both commercial and diplomatic exchanges during the second half of the first millennium. The Srivijayan polity in Southeast Asia, for example, established strategic relations with the powerful Chola court in south India and at the same time maintained strong commercial connections with economically vibrant Song China. The Srivijayan ruler based in Sumatra exerted significant control over maritime commerce taking place through the Straits of Malacca. Srivijaya not only became the main transit center for commodities destined for or originating from the markets in China but also regulated ships passing through the

Straits of Malacca. Srivijaya's monopoly over maritime commerce with China was challenged in 1025 by the Chola court, which reportedly launched a massive naval attack on the Srivijayan ports. This attack signified the beginning of the militarization of the Indian Ocean world, something that was not possible without the development of navigational and shipbuilding technologies.[30]

New oceangoing ships and methods of way finding also had a significant impact on the nature and volume of commodities traded across the Eurasian world. Bulk goods including incense, aromatics, and spices became more commonly traded items as tonnage increased in the large ships. Frankincense from the Persian Gulf, pepper from southern India, and sandalwood from Southeast Asia were frequently traded across the Indian Ocean.[31] Similarly, porcelain from China was exported in larger quantities as ships, rather than camels, provided a safer mode of transportation over long distances.[32] Maritime exchanges continued to expand between the thirteenth and fifteenth centuries, at first overlapping with intense overland interactions that occurred as Mongol empires formed across the continent, a time when a Tuscan commercial center such as Siena existed on "the periphery of Mongol Eurasia."[33] Indeed, during the so-called pax Mongolica, maritime and overland routes became more intimately intertwined. This is evident from the travels of Marco Polo and Ibn Battuta, the extensive knowledge about the "world" collected by the Persian historian Rashīd al-Dīn, and the expansive dispersion of Chinese porcelain ware. As individuals and whole groups of peoples moved across Eurasia, their practices and ideas traveled with them. These long-distance movements of peoples, including the forced migrations of craftspeople under the Mongols, and informal exchanges such as the capture of prisoners in war caused sociotechnical systems—from weaving to weapons—to move along with them.

During the Mongol period ships from Yuan China began to dominate the eastern Indian Ocean networks. These Chinese ships made frequent calls at the Malabar coast of India, and Chinese diasporic communities formed during this period in Southeast Asian ports.[34] These communities became more active in the early fifteenth century, when the Ming court instituted strict restrictions on foreign trade. The Ming court also launched seven massive naval expeditions led by the commander Zheng He between 1405 and 1433. These expeditions resulted in the formation of new hubs of maritime commerce including Malacca in Southeast Asia, Cochin in southern India, and Malindi in eastern Africa. The voyages, which included several hundred ships, over twenty thousand soldiers, and advanced gunpowder weapons, also had a significant impact on the movement of people and animals, at the same time as they militarized the oceanic space and consolidated state participation in maritime trade. Only a half century later, a new group entered into these already well-established connective pathways, and these early European trading companies, especially the Portuguese, imitated many points of strategy that had informed Zheng He's voyages, including control of major choke points and attempts to monopolize private trading activity.[35]

The Global Integration of Eurasia

The integration of Eurasia through commerce and religious transmission peaked during the colonial period as commodities such as tea, spices, and textiles moved in large quantities within the region. Active missionary work by Jesuits and members of other Christian churches connected regions as distant as Rome, Goa, Macao, Malacca, and Kagoshima. Similar to Buddhist and Muslim proselytizers, these missionaries also brought with them unique ideas of observing the cosmos, curing diseases, and technical expertise. In the *longue durée* of connections and interactions across Eurasia, the European traders and missionaries were just the latest contributors to integrate a vast space that had transcended imagined continental divides far into prehistory.[36] More importantly, in the sixteenth and seventeenth centuries, the Eurasian world became integrated into the circulations across the Pacific and Atlantic oceans, resulting in new forms and more intensive circulations of people, objects, and ideas.

The web of exchange, highlighted by the foregoing survey, that knit together Eurasia has often been confusingly called the "Silk Road." However, it could be named the Jade or Horse Road, and, from other regional perspectives, the rhubarb and paper routes.[37] The concept of a singular Silk Road has also masked an equally ancient "maritime silk route," and an even earlier eastward flow of furs along a northerly "fur route," as well as a "musk route" connecting Tibet and the Islamic world from the eighth century. But, most significantly, as this list of alternative names for the route implies, it has masked the very nature of human interaction across the continent (figure 2.3).[38] Exchange across regional and political boundaries was the norm. Indeed, such flows were simply extended further east and west around the globe as the Americas were forcibly incorporated within the Afro-Eurasian ecumene from the sixteenth century on.

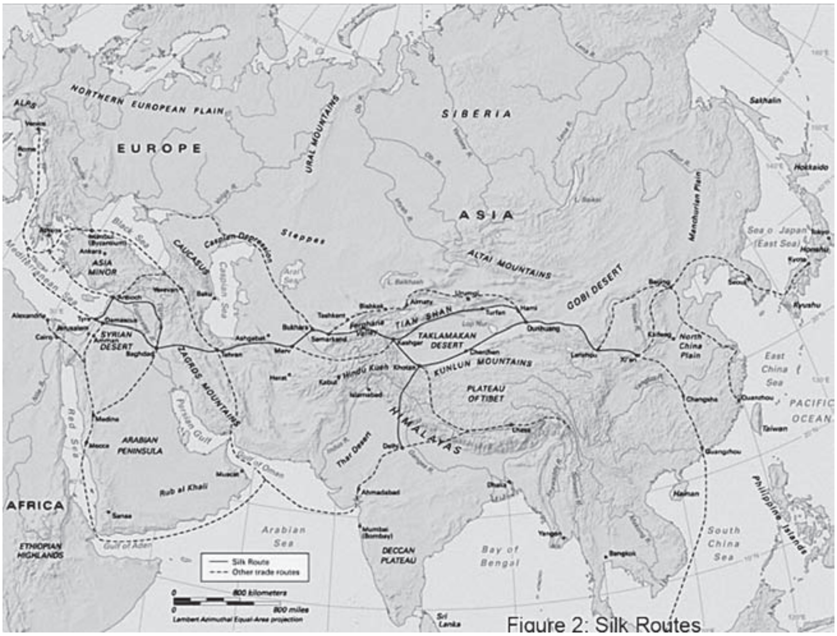

Figure 2.3. Silk routes. Trade routes in the Middle Ages.

By following these flows of materials, we can discern some general patterns in the movement of materials, practices, and knowledge. Some moved with particular speed, especially those related to the techniques of survival, both of sustenance and of war (copper alloys, the stirrup, gunpowder, to name just three). Although our primary documentary evidence for the movement of foodstuffs comes from the period when New World foods entered into already well-established trade routes at the eastern end of Eurasia, a few examples indicate how fast such crops could travel. The peanut cultivated in the Chaco region of South America, for example, could already be found growing as a food crop near Shanghai by the 1560s or 1570s at the latest; sweet potatoes had arrived in China at least by the 1560s; New World maize was established in China in 1555, where it had been brought by Turkic frontiersmen, and along the Euphrates by 1574.[39] Some of the first European representations of New World plants, maize corn and hot peppers, appeared in German herbals (in particular that of Leonhart Fuchs in 1543) as plants from the eastern ends of Eurasia, with maize called "Turkish grain" and chilis labeled "Calcutta peppers" (figure 2.4). This may indicate that these New World plants entered Europe from easterly ports as well as from New Spain and Lisbon, for maize may have been cultivated on a larger scale first in the Ottoman Empire and only later in Europe. We know that it was already being grown in Spain on a small scale in 1530, only a few years after it had first been brought to Europe. In the Ottoman Empire, maize was called *Mısır buğdayı* (Egyptian wheat) as it was first introduced into Egypt sometime after the Turkish conquest in 1517 and later reached Anatolia and Istanbul via Syria.[40] As for "Calcutta peppers," the Portuguese took New World peppers to India, where they came to be cultivated and regarded as native and then flowed west as one component of "curry." Whatever the source of Fuchs's plant names, it is clear that food plants traveled in advance of recorded contact between peoples, carried by sailors and other anonymous intermediaries (figure 2.5). Their cultivation in new soils must have occasioned much experimentation by their growers both in the field and at the dinner table. As in the case of "Turkish corn," the very name of the material at times indicated this movement. Take, for example, the words "china" in English for porcelain and *cini* in Turkish for decorated ceramics, or the north Indian word *cini* ("Chinese") for sugar, which also seems to have derived from local reference to that ubiquitous trade object porcelain.[41] In the case of ceramics produced in Iznik in northwestern Anatolia, called *Iznik* çinileri (Iznik chinoiserie), a mix of Ottoman and Chinese decorative elements, the name itself reveals a long itinerary of this "material complex" from East Asia through Central Asia, via Persia, to Anatolia. Similarly, the instruments of war and the trappings, ceremonies, and techniques of staging power, such as the royal hunt carried out with noble animals and raptor birds, also spread extensively throughout Afro-Eurasia (figure 2.6).[42] Techniques of governing may also have flowed across this space: Roger II (r. 1132–1154) of Norman Sicily gathered information and informants from all parts of the world, and his plan for training members of the court in three stages of education may have been based upon his knowledge of Chinese practices.[43] Common mathematical techniques for dividing up partnerships and profits for the purposes of commerce traveled far and wide, as well as the problems of calendar making, and the tables needed to construct star charts, astrological horoscopes, and for figuring the direction of Mecca. Story problems for teaching rules of calculating the relative speeds of animals and planets chasing one another are found in China in the first century, in India in the seventh century, and in Latin—in their simplest form—in the ninth.[44]

Shared patterns of consumption emerged as well, Christopher Bayly has argued, that constructed the value of unique, rare, and unusual objects, part

Figure 2.4. Calechutischer Pfeffer (Calcutta pepper). Leonhart Fuchs, *New Kreüterbuch* (1543). Signatur 19 257 4°, Stadtbibliothek Ulm.

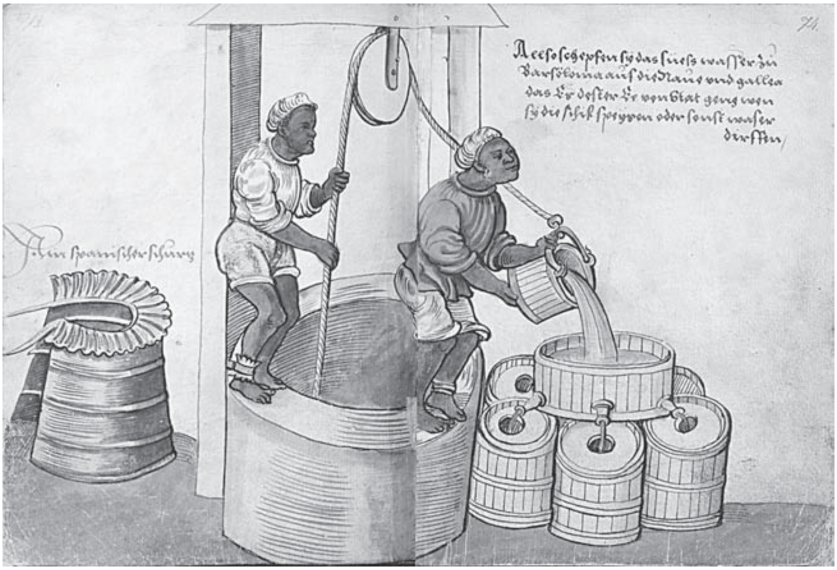

Figure 2.5. Provisioning of water for ships by slaves in Barcelona. Christoph Weiditz, "Trachtenbuch" (1529). Fols. 73–74, Hs. 22474, Germanisches Nationalmuseum Nürnberg. Germanisches Nationalmuseum, digital library. Public domain.

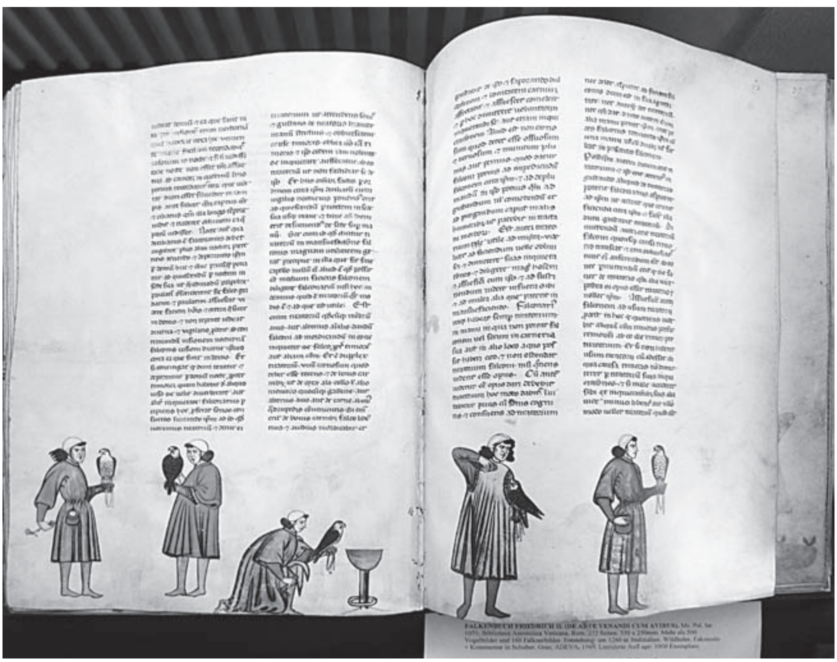

Figure 2.6. European traces of the royal hunt. Frederick II, Hohenstaufen, *De arte venandi cum avibus* (thirteenth century). Vatican, Ms. pal. lat. 1071.

of a material imaginary that could make real and tangible qualities of distant parts of a realm. The simultaneously social, moral, and exchange economy in which such valuable wondrous objects were embedded extended downward and outward through the courts of regional nobles (figures 2.7 and 2.8). At the same time the consolidation of the major religions—with their similar practices of prayer, sacrifice, and pilgrimage by which devotees pursued the signs and sites of gods scattered throughout the world—brought into existence a large infrastructure of transport, food, credit, and international trade to support their practices of devotion and their extensive systems of pilgrimage (spice for incense and Brahminical food restrictions, for example, and salted fish for Catholic fasting).[45] The individuals traversing these various pilgrimage and trade routes shared a common preoccupation with the relation of body and spirit—which sometimes converged in similar views of health and a common range of medicines, even when understood differently. Historians have debated the similarities and translatability of central medical concepts—such as "life-breath" and "winds" (*qi*, *pneuma*, and *prāna*) that can be found in ancient Chinese, Greco-Roman, and South Asian medical writings as well as the Greco-Roman-Islamic system of the four humors and the Chinese system of yin/yang. Some historians have argued that all these systems have origins in a Eurasia-wide notion of balance and the tempering of contrary forces in order to bring the body to a healthy state.[46] Historians have also debated what role these conceptual systems played in the variety of medical practices and the often similar substances and remedies that informed care of the body throughout Eurasia. At the most general level, an individual's constitution was viewed as influenced by the heavenly bodies, and many of the materials by which bodies could be tempered and balanced were transported over very long distances via the same trade routes as were traveled by weapons and rarities. Spices (wholly integrated into the health framework) and all manner of materia medica—including medicinal plants (such as dragon's blood), animal parts (such as rhinoceros horn), precious stones (for example, turquoise for eyesight and red coral to stop blood flow), and mineral-based concoctions (such as large quantities of sulfur, which flowed by the ton into Tang China from Indonesia and Japan between the seventh and tenth centuries CE, used both for industrial processes and fireworks and to temper the constitution)—all possessed medicinal virtues and commanded high exchange values throughout Eurasia.[47]

A literature on "wonders," which began to codify and canonize the marvelous qualities of these objects and simultaneously to stimulate desire for

Figure 2.7. Mother of pearl and inlay (Gujarat, India, sixteenth century), casket with northern European mountings. Staatliche Kunstsammlungen Dresden. Photo: Jürgen Karpinski.

them, grew up around the exchange in rarities and medicines described above. In this literature India was identified as the primary location of wonders and marvels, as well as the source of precious stones, a role that, along with Sri Lanka, it had played at least since 400 BCE.[48] While the literature on wonders extends back even to Greek texts, the view that wonders originated in India, Berthold Laufer argues, emerged out of an aristocratic Arabic-Persian milieu.[49] In Europe the very first of 129 wonders that Gervase of Tilbury lists in his *Otia imperialia* (1210) is the magnet, "an Indian stone," and William of Auvergne (ca. 1180?–1249) regarded "parts of India and other adjoining regions" as containing "a great quantity of things of this sort, and on account of this, natural magic particularly flourishes there."[50] Muzaffar Alam and Sanjay Subrahmanyam note that medieval travel literature throughout the world includes "wonders" or "aja'ib" (in other words, marvels, astonishing things or "mirabilia"), most often told in first-person narrative, like that of tenth-century Persian author writing in Arabic, Buzurg ibn Shahriyār, in *Kitab 'ajāyib al-Hind* (The wonders of India, ca. 960), which recounts marvels of all sorts purportedly told to him on board merchant ships, including many involving fearsome snakes, future-divining lizards, and crayfish that, for example, fell into the Sea of Senf, turned into stones, and were subsequently "carried into Iraq . . . and used in making an ointment to cure eye complaints."[51] Such travelers' reports and sailors' yarns (best known perhaps in those of Sinbad the Sailor and the *Arabian Nights*) spread widely through the "Maritime Silk Route," forming the stuff of "folk tales" across Eurasia.[52] On land, troubadours "blended traces of Sufi and Islamic mysticism with Manichaean Christian narratives," as in the story of Barlaam and Josaphat, which tells the tale of a prince's renunciation of all worldly pleasures and material belongings. This version of a Manichaean tale began as a recounting of the life of the Buddha, but it emerged again in the eleventh and twelfth centuries in Islamicate and Christian lands, apparently carrying other Manichaean beliefs along with it, and perhaps even contributing to the Manichaean revolt in the Northern Song in the twelfth century and the Cathar crisis in Languedoc.[53]

Even this brief and cursory survey of the pathways of exchange that coursed across Eurasia since prehistoric times shows that the flows of goods and of the peoples who mined, sourced, worked, produced, transported, and consumed them helped to constitute the "material-knowledge complexes" that both structured and fostered this exchange. In a reciprocal process between humans and their material environment, the mundane production of material things informed and transformed theoretical formulations of knowledge. The transfer of practical techniques across space fostered new ways of controlling both nature and other people. The movement of goods and their trade gave rise to people telling oral tales and forming and reforming conceptual, spatial, and intellectual imaginaries, some of which were compiled, written down (only sometimes by identifiable authors), and translated into various vernacular and learned languages in texts that also moved across great distances, where they stimulated the itineraries of yet further travelers, objects, and ideas.

Figure 2.8. Interior accoutrements of casket, created from Indian Ocean naturalia and set in northern European mountings. Staatliche Kunstsammlungen Dresden. Photo: Jürgen Karpinski.